CBSE Practice Worksheet

Class 8 | Mathematics | Difficulty: Hard | Set D

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$
   - (A) $x - 2$
   - (B) $x + 2$
   - (C) $(x-2)^2$
   - (D) $x^2 - 4$
2. Solve: $\sqrt{(x + 7)} = x - 5$
   - (A) $x = 9$
   - (B) $x = 2$
   - (C) $x = 9$ or 2
   - (D) No solution
3. Simplify: $(x^3 - 8)/(x - 2)$
   - (A) $x^2 + 2x + 4$
   - (B) $x^2 - 2x + 4$
   - (C) $x^2 + 4$
   - (D) $x^2 - 4$
4. If a:b = 3:4 and b:c = 5:6, find a:c
   - (A) 3:6
   - (B) 5:8
   - (C) 15:24
   - (D) 1:2
5. If $x + 1/x = 5$, find $x^2 + 1/x^2$
   - (A) 21
   - (B) 23
   - (C) 25
   - (D) 27
6. Solve: $3(x-2)/4 = (2x+1)/3$
   - (A) $x = 14$
   - (B) $x = 22$
   - (C) $x = 26$
   - (D) $x = 30$
7. Find: $(-1)^{1}$■■ + (-1)■■
   - (A) 0
   - (B) 2
   - (C) -2
   - (D) 1
8. What is the cube root of -125?
   - (A) 5
   - (B) -5
   - (C) ±5

(D) undefined

9. Factorize: $x^2 - 5x - 14$
   (A) (x-7)(x+2)
   (B) (x+7)(x-2)
   (C) (x-7)(x-2)
   (D) (x+7)(x+2)

10. If 2■ = 3■ = 6■, which is true?
   (A) 1/z = 1/x + 1/y
   (B) z = x + y
   (C) z = xy
   (D) z = x - y

Section B: Fill in the Blanks (1 mark each)

11. If 5■ = 125, then x = ___

12. $^3\sqrt{(-64)}$ = ___

13. Rationalize: $1/(\sqrt{3} + 1) = (\sqrt{3} - 1)/$___

14. If $x^2 - 6x + 8 = 0$, then x = ___ or ___

15. If x + y = 7 and xy = 12, then $x^2 + y^2$ = ___

16. If a + b + c = 0, then $a^3 + b^3 + c^3$ = ___

17. If log x = 2, then x = ___

18. $(x + 1/x)^2 - (x - 1/x)^2$ = ___

Section C: Short Answer Questions (2 marks each)

19. Find the value of x: 3^(2x-1) = 27^(x-2)

20. Simplify: (x■ - 16)/(x^2 + 4)

21. Solve: $|x - 3| + |x + 2| = 7$

22. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

23. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

24. Solve: $x^2 - 7x + 12 = 0$

25. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

--- End of Worksheet ---